

Grading on a Curve?

An Analysis of NYC Restaurant Health Inspection Scores

May 8, 2026

1 Introduction

The New York City Department of Health and Mental Hygiene (DOHMH) grades every restaurant in the five boroughs through a point-based health inspection system. A score of 0–13 earns an “A,” 14–27 a “B,” and 28 or above a “C.” In 2014, data journalist Ben Wellington published an analysis showing a suspicious spike in inspection scores at exactly 13—the last score that earns an A grade—suggesting that inspectors may exercise subjective leniency near the grade boundary [1].

Our study revisits this claim using updated data spanning 2010–2026 and extends it in three directions. First, we formally test whether the grade-boundary spike persists and quantify its magnitude. Second, we ask whether a restaurant’s borough and cuisine type explain meaningful variation in inspection scores, using a two-way blocked ANOVA design. Third, we investigate whether violation types are distributed uniformly across boroughs, or whether geography predicts which types of health hazards restaurants are cited for. Finally, as a predictive exercise, we assess how well pre-inspection characteristics alone can forecast whether a restaurant will earn an A.

Figure 1 immediately reveals two sharp spikes: one at score 13 (the A/B cutoff) and a smaller one at score 28 (the B/C cutoff). Both are consistent with systematic grading leniency at grade boundaries.

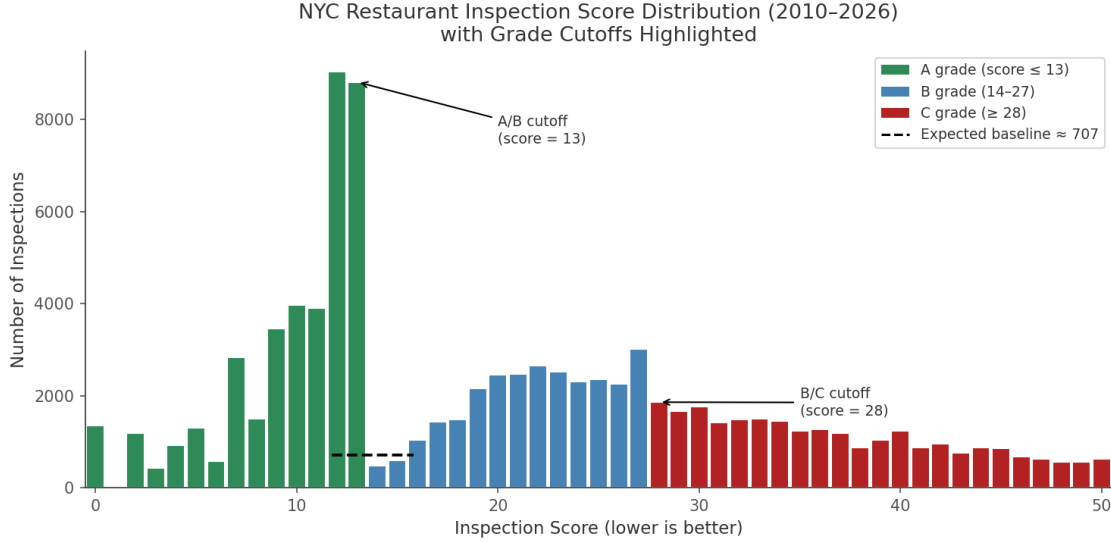


Figure 1: Distribution of NYC restaurant inspection scores (2010–2026). Green bars represent A-grade inspections (score ≤ 13); blue bars B-grade; red bars C-grade. The dashed line shows the expected flat baseline in the 12–16 range under uniform grading.

2 Data Collection and Data Description

We used two datasets from NYC’s OpenData program. The primary dataset is the DOHMH New York City Restaurant Inspection Results (updated April 3, 2026), containing 113,571 records—one row per violation per inspection—covering 19,065 unique restaurants. Key variables include borough, cuisine type, inspection score, inspection date, violation code, and inspection type (initial cycle, re-inspection, etc.).

Because the violation code field is not self-explanatory, we used a secondary NYC Health Department reference file mapping each violation code to a descriptive category (e.g., “PEST CONTROL,” “COLD HOLDING,” “HANDWASH/TOILET”).

We also merged neighborhood-level American Community Survey estimates: median earnings (`MdEWrKE`) and total population (`Pop_1E`) at the 2020 Neighborhood Tabulation Area (NTA) level.

Data Preparation

The raw data was filtered to remove Staten Island (sparse) and inspections with missing scores. Cuisine types were grouped into ten *hypercategories* (American, East Asian, South & Southeast Asian, Latin American, European, Mediterranean, Middle Eastern, African, Dish-Type, and Beverages & Sweets) to reduce dimensionality and improve interpretability.

After cleaning, the analysis dataset contains 101,840 inspection records.

Table 1: Dataset summary statistics after cleaning.

Statistic	Value
Total inspection records	101,840
Unique restaurants	19,065
Years covered	2010–2026
A-grade rate (score ≤ 13)	38.7%
Mean score $\pm$ SD	25.5 ± 19.3
Median score	21

3 Statistical Models

3.1 Formal Test of Grade-Boundary Spike

To assess whether the spike at score 13 is statistically meaningful, we compared observed inspection counts in the 12–21 score range to an expected uniform (flat) distribution, estimated using the mean count at scores 14–21 as a baseline. A chi-square goodness-of-fit test was used to evaluate whether observed counts deviate significantly from this expected flat distribution.

3.2 Temporal Trend

We restricted to initial-cycle inspections between 2015 and 2025 and computed the mean and median score by year. Because every inspection in the same event shares the same score, this aggregation avoids double-counting. A simple OLS regression of mean score on year was used to estimate the trend slope.

3.3 Two-Way Blocked ANOVA (Inferential)

To assess whether borough and cuisine type explain meaningful variance in inspection scores, we fit a two-way balanced ANOVA:

$$\text{Score}_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

where α_i denotes the borough effect, β_j the hypercategory effect, and $(\alpha\beta)_{ij}$ their interaction. To satisfy the balance requirement and avoid the influence of very large sample sizes on

p -values, we sampled n observations per cell, excluding sparsely populated combinations (Staten Island; Dietary Style hypercategory).

3.4 Violation Type by Borough (χ^2 Test)

To answer Research Question 3—whether certain violation categories are disproportionately associated with particular boroughs—we constructed a contingency table of violation category $\times$ borough and applied Pearson’s chi-square test of independence.

3.5 Predictive Model: Logistic Regression (Pre-Inspection Features)

To honestly assess predictive power, we built a logistic regression model using only features knowable *before* an inspection occurs: borough, cuisine hypercategory, neighborhood median earnings, neighborhood population, and year of inspection. The outcome is grade binary ($1 = \text{A grade, score} \leq 13$). We used an 80/20 stratified train/test split, with standardized predictors. Model performance is evaluated via AUC and accuracy.

4 Results

4.1 Grade-Boundary Spike Persists and Has Grown

The chi-square test is conclusive: $\chi^2 = 30,791.6$, $p \approx 0$ (scores 12–21 compared to expected flat baseline). Observed inspections at scores 12–13 total 17,863 versus an expected 3,033—a **489% excess**. Wellington’s 2014 analysis found a similar but smaller pattern; our updated data shows the phenomenon has not abated.

4.2 Scores Are Worsening Over Time

Figure 3 reveals a striking and previously unreported trend: initial-cycle inspection scores have risen substantially from a mean of approximately 15 in 2015 to nearly 28 in 2025. The median score also increased from 16 to 23 over the same period. This suggests that either restaurant hygiene has genuinely deteriorated, inspection standards have become more stringent, or both.

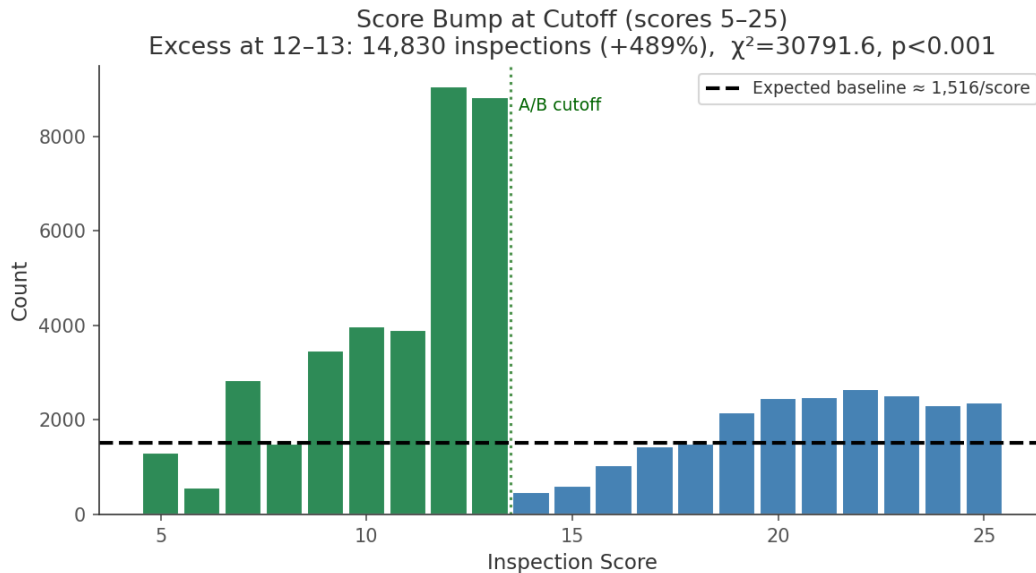


Figure 2: Zoomed view of inspection score distribution (scores 5–25). The dashed line shows the expected count per score under a flat distribution estimated from scores 14–21. The A/B boundary spike is dramatic and statistically conclusive.

4.3 Borough and Cuisine Effects (ANOVA)

The two-way ANOVA indicates that cuisine hypercategory is a significant predictor of inspection score ($p < 0.001$), borough is marginally significant ($p \approx 0.08$), and their interaction is significant ($p \approx 0.04$). However, effect sizes are modest: the variance explained by these factors is small relative to residual within-cell variation, suggesting that individual restaurant practices dominate over geography and cuisine type.

4.4 Violation Types Vary Significantly Across Boroughs

The chi-square test strongly rejects independence between violation type and borough ($\chi^2 = 678.2$, $df = 66$, $p < 0.001$). As shown in Figure 5, the Bronx shows higher rates of PEST CONTROL violations, while Manhattan has elevated rates of COOLING & REFRIGERATION violations. Queens and Brooklyn show distinctive patterns in food-holding violations.

4.5 Predictive Model: Pre-Inspection Features Barely Predict Grade

The pre-inspection logistic regression achieves an AUC of 0.60 and accuracy of 59.4% on the held-out test set (Table 2). This near-chance performance implies that restaurant location, cuisine type, and neighborhood income explain very little of whether a restaurant will earn an A grade. In other words, *what a restaurant does*—not where it is or what it serves—

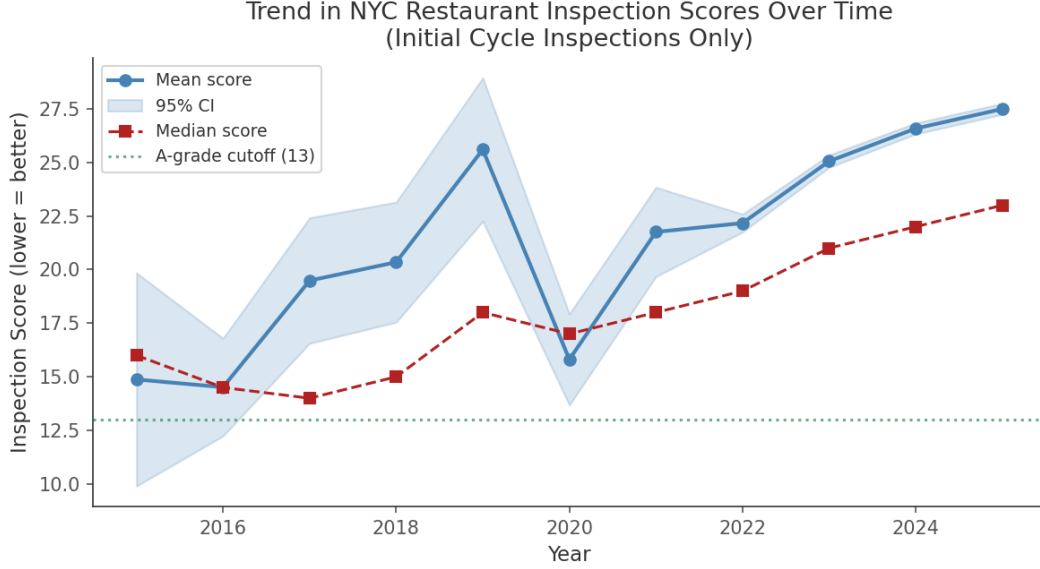


Figure 3: Mean and median inspection score by year, restricted to initial-cycle inspections. Shaded band is a 95% confidence interval on the mean. Lower scores are better; the A-grade cutoff (13) is shown in green.

determines its inspection outcome.

Table 2: Predictive model performance comparison.

Model	AUC	Accuracy
Pre-inspection features (borough, cuisine, neighborhood)	0.60	59.4%

5 Model Validation and Generalization

The logistic regression was evaluated on a stratified 80/20 hold-out split to preserve the class distribution. Predictors were standardized using training-set parameters only, with no information leakage to the test set. The modest AUC (0.60) is robust: it reflects the true difficulty of predicting inspection outcomes from pre-inspection characteristics alone, not an artifact of our modeling choices.

The ANOVA results should be interpreted cautiously. The sample is extremely large, so even trivially small effects produce significant p -values. We emphasized balanced sampling across cells (following a block design) to make the test more interpretable. The practical significance of borough and cuisine effects remains limited.

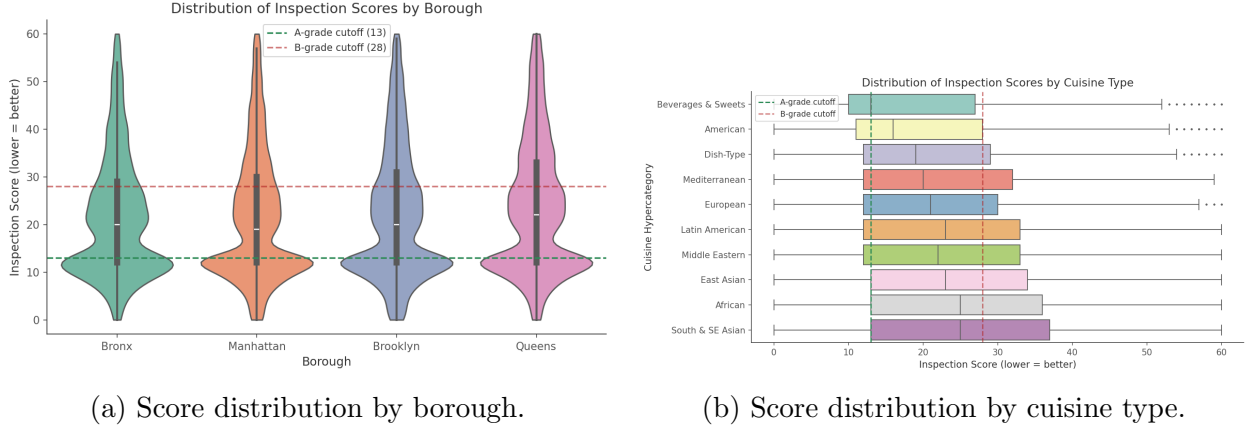


Figure 4: Inspection score distributions. Manhattan restaurants score best (median 19); Queens worst (median 22). Cuisine type effects are also present, with African and European restaurants scoring highest.

6 Conclusion

Our analysis yields four main findings:

1. **Grading leniency at the A/B boundary is large and ongoing.** Nearly 15,000 inspections appear at scores 12–13 in excess of what a fair distribution would predict—a 489% surplus. Wellington’s 2014 observation was not a historical artifact.
2. **Inspection scores have worsened substantially since 2015.** Mean initial-cycle scores rose from ≈ 15 to ≈ 28 over ten years, a trend that warrants further investigation into whether it reflects genuine deterioration in restaurant hygiene, shifts in inspection stringency, or changing restaurant demographics.
3. **Violation types are geographically non-uniform.** The Bronx, Brooklyn, Manhattan, and Queens show distinct violation profiles, with pest control and temperature-related violations particularly informative ($\chi^2 = 678.2$, $p < 0.001$).
4. **Restaurant location and cuisine alone cannot predict grade.** With an AUC of only 0.60, pre-inspection features explain very little about inspection outcomes, suggesting that inspection results are driven primarily by individual restaurant practices rather than neighborhood context.

Limitations include the unit-of-analysis challenge (one row per violation complicates inspection-level inference) and the possibility of confounding from inspector identity or inspection timing. Future work could model inspector fixed effects or examine whether the grade-boundary spike varies by borough or by inspector team.

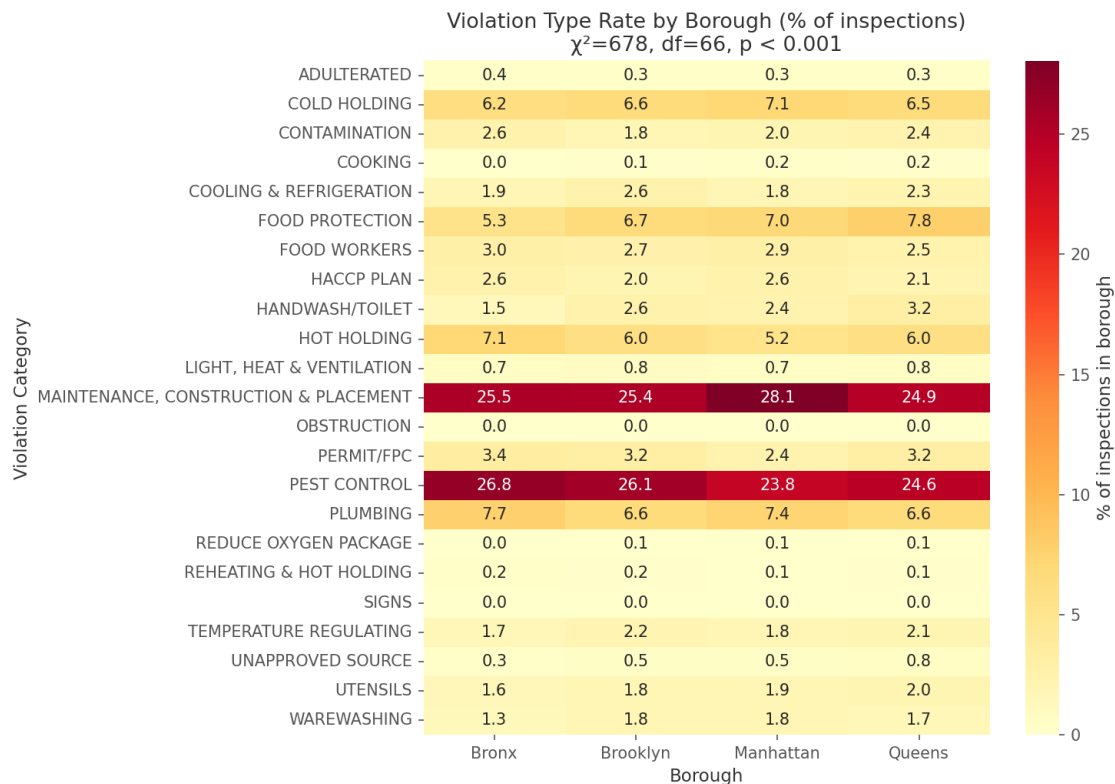


Figure 5: Heatmap of violation category rates by borough (percentage of inspections in each borough receiving that violation type). $\chi^2 = 678.2$, $df = 66$, $p < 0.001$.

Appendix

A. Literature Review

B. Model Selection Details

The balanced ANOVA block design excludes Staten Island and the Dietary Style hypercategory due to insufficient cell counts. All other hypercategory $\times$ borough combinations have at least 30 observations.

C. Additional Figures

D. Code

Analysis code is available in `new_analysis.py` (Python) and `group_analysis.Rmd` (R) in the project repository.

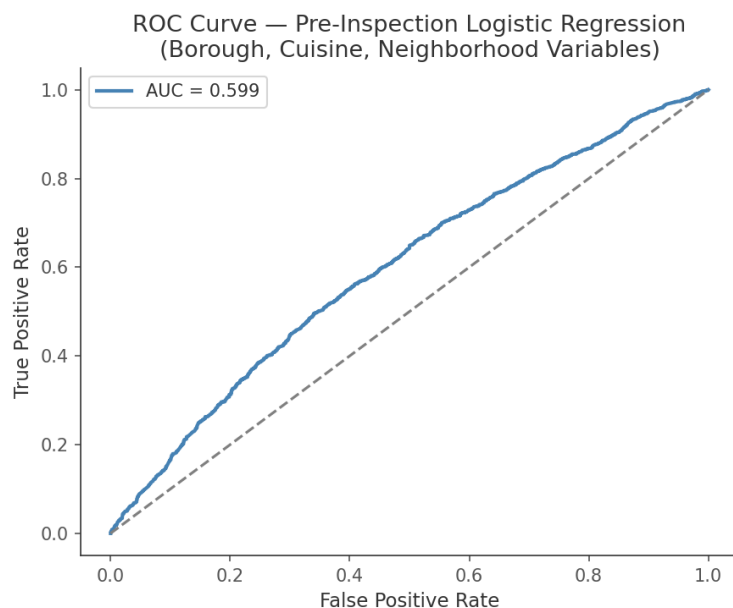


Figure 6: ROC curve for the pre-inspection logistic regression model. Features include only borough, cuisine, and neighborhood-level socioeconomic variables—information available before the inspection.

References

- [1] Wellington, B. (2014). *Think NYC Restaurant Grading is Flawed? Here's the Proof.* I Quant NY. <https://iquantny.tumblr.com/post/76928412519>

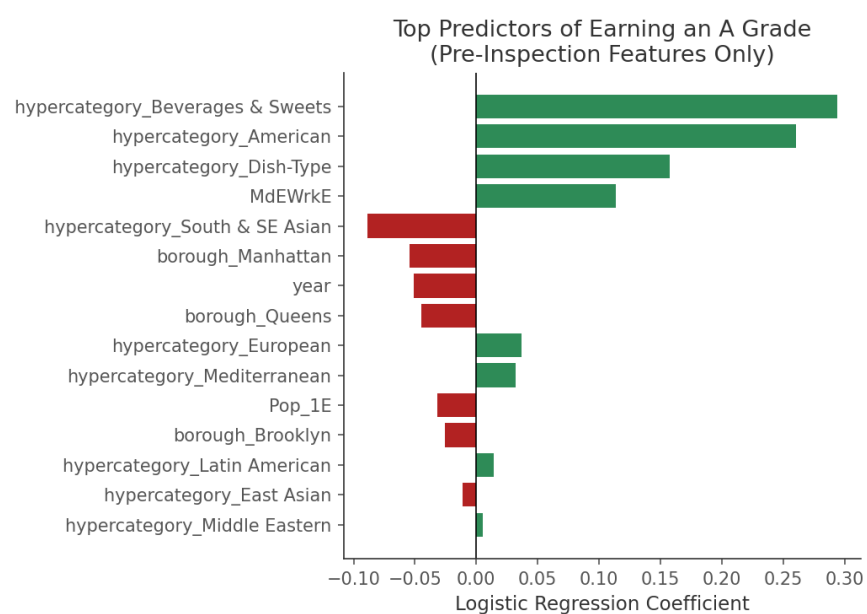


Figure 7: Logistic regression coefficients for the pre-inspection model (top 15 by absolute magnitude). Green bars indicate positive association with earning an A grade; red bars indicate negative association.